

**Design of Bifunctional Fentanyl Analogs: Dual-targeting the Mu-Opioid and
Nociceptin Receptors for Safer Analgesia**

**Jon Baldursson
Sunset High School
Portland, Oregon, United States**

Abstract

The opioid crisis remains a major public health challenge in the United States, driven largely by the widespread misuse of fentanyl and its analogs. Fentanyl is a potent synthetic opioid whose primary mechanism of action—activation of the Mu-Opioid Receptor (MOR)—induces life-threatening respiratory depression and high addiction potential through downstream dopaminergic signaling. Despite its clinical utility, no safer alternative of equivalent analgesic potency currently exists. This project aimed to address that gap by designing, synthesizing, and validating novel bifunctional fentanyl analogs that co-activate both the Mu-Opioid Receptor (MOR) and the Nociceptin/Orphanin FQ receptor (NOP), leveraging the established modulatory role of NOP on opioid side-effects.

A fragment-based computational workflow was used to identify candidate functional groups from known NOP-active compounds, which were then screened against the ZINC-22 chemical database using a multi-parameter similarity equation. Fifty-seven candidate analogs were prepared in silico as 3D conformers and docked to the prepared MOR and NOP crystal structures using AutoDock Vina, filtering for dual binding affinity below -9.0 kcal/mol. The four top-ranking scaffolds were subjected to molecular dynamics (MD) simulations using OpenMM. The naphthyl-modified scaffold demonstrated the lowest root-mean-square deviation (RMSD) at both receptors and was selected for synthesis.

Synthesis of Naphthyl-Norfentanyl was performed via reductive amination at OHSU, yielding the compound at 94.8% purity, confirmed by LC-MS and H-NMR. In-vitro validation using HEK293 cAMP assays revealed an IC₅₀ of 5 nM at MOR (comparable to fentanyl's 3 nM) and 11 nM at NOP (superior to the reference agonist Ro64-6198 at 20 nM). In dual MOR/NOP co-expressing cells, the compound produced a maximal cAMP inhibition (E_{max}) of only 55% versus 93% for fentanyl alone, confirming active NOP-mediated attenuation of MOR signaling. Brain slice GIRK electrophysiology further demonstrated approximately 2-fold lower peak currents compared to fentanyl ($p < 0.001$). These findings support the hypothesis that bifunctional MOR/NOP agonism can preserve analgesic potency while substantially reducing respiratory depression risk, offering a promising scaffold for next-generation safer opioids.

Introduction and Goal

Background: The Opioid Crisis and the Role of Fentanyl

The opioid epidemic represents one of the most severe public health crises in modern American history. Fueled in large part by the proliferation of illicit synthetic opioids, particularly fentanyl, overdose fatalities have reached unprecedented levels over the past decade. Oregon has been disproportionately affected: between 2019 and 2023, the state experienced a 1,500% increase in fentanyl-related overdose deaths, the highest proportional increase in the United States (Oregon Health Authority, 2024). Growing up in Portland, the researcher observed firsthand the devastating community impact of this crisis, motivating an investigation into safer pharmacological alternatives.

Fentanyl is a fully synthetic opioid approximately 100 times more potent than morphine and heroin. Its clinical utility stems from its rapid onset, high analgesic efficacy, and well-characterized pharmacokinetic profile. Fentanyl exerts its effects by binding to the Mu-Opioid Receptor (MOR), a G-protein-coupled receptor (GPCR) expressed throughout the

central nervous system, primarily via interaction with the Asp147 residue in the receptor binding pocket. However, this same mechanism underlies its most dangerous properties.

MOR activation in the PreBötzinger Complex (preBötC), a respiratory rhythm-generating nucleus in the brainstem, suppresses normal breathing control and can induce fatal respiratory depression. Separately, MOR activation in the Ventral Tegmental Area (VTA) disinhibits dopaminergic neurons projecting to the nucleus accumbens, producing the reward and euphoric effects responsible for fentanyl's high addiction potential. Together, these mechanisms create a compound that is simultaneously highly effective as a painkiller and highly dangerous when misused or accidentally overdosed.

Existing Alternatives and Their Limitations

Current alternatives to fentanyl fall into two broad categories: other opioids and non-opioid analgesics. Morphine offers a well-established clinical track record and remains widely available, but it is significantly less potent than fentanyl, has a slower onset of action, and, critically, it retains the same fundamental risks of respiratory depression and physical dependence, since it also acts primarily through MOR. Non-opioid analgesics such as acetaminophen and ibuprofen are safe and broadly accessible, but their analgesic ceiling is far below what is achievable with opioids, making them inappropriate for moderate-to-severe acute or post-surgical pain.

Among existing treatments for opioid overdose, naloxone (Narcan) remains the primary emergency intervention. Naloxone reverses an overdose by competitively blocking MOR, rapidly displacing fentanyl, and restoring respiratory function. However, it is purely symptomatic and does not alter the intrinsic risk profile of opioids themselves. Furthermore, naloxone does not affect the dopaminergic reward pathway, meaning it does not address addiction potential. The critical gap in the landscape is therefore a compound that preserves fentanyl's therapeutic potency while intrinsically limiting its capacity to cause respiratory depression and dependence.

The Nociceptin/Orphanin FQ Receptor as a Modulator of Opioid Side-Effects

Recent pharmacological research has highlighted the Nociceptin/Orphanin FQ receptor (NOP), also designated as the ORL1 receptor, as a compelling target for addressing the side-effect profile of classic opioids. Like MOR, NOP is a Gi-protein-coupled receptor distributed throughout the central nervous system. Its endogenous ligand, nociceptin (N/OFQ), modulates pain, stress responses, and reward circuitry. Critically, NOP activation has been shown to oppose and attenuate the same downstream signaling pathways that MOR overstimulates. Specifically, NOP activation in the VTA suppresses dopaminergic neurotransmission, counteracting the reward signaling induced by MOR activation. Similarly, NOP-mediated modulation of brainstem respiratory centers appears to partially counteract opioid-induced respiratory depression (Ciccocioppo et al., 2025).

This regulatory relationship suggests that a compound capable of simultaneously activating both MOR and NOP—a bifunctional or dual agonist—could, in principle, achieve strong analgesia through MOR while NOP co-activation limits the ceiling of MOR-driven side-effects. This concept, sometimes termed NOP/MOP dual agonism, has garnered interest as a therapeutic strategy. Cebranopadol, a clinical-stage compound with mixed MOR/NOP agonism, has demonstrated analgesic efficacy with a potentially improved safety profile in early trials, lending further support to this approach.

Research Goal and Hypothesis

The overarching goal of this project was to design, synthesize, and validate novel fentanyl-derived bifunctional analogs that co-activate MOR and NOP, to produce compounds that retain fentanyl-level analgesic potency while demonstrating reduced respiratory depression and addiction potential indicators in vitro.

Rather than designing entirely new chemical entities from scratch—an approach that is prohibitively slow, expensive, and high-risk—this project hypothesized that modification of the existing norfentanyl scaffold through substitution at the piperidine nitrogen could produce fentanyl-adjacent structures with meaningful NOP affinity. This strategy preserves the well-validated pharmacokinetic properties and synthetic accessibility of the fentanyl scaffold while selectively introducing chemical features associated with NOP binding. The central hypothesis was: substitution of the phenethyl group on norfentanyl's piperidine nitrogen with NOP-activity-associated functional groups will produce bifunctional analogs with dual MOR/NOP agonist activity, and NOP co-activation will demonstrably reduce the ceiling of MOR-driven signaling in co-expressing cellular systems.

Methods

Fragment Identification and Analog Library Generation

Functional group candidates for substitution at the norfentanyl piperidine nitrogen were identified through a two-stage cheminformatics workflow. In the first stage, a set of known NOP agonists—including cebranopadol, Ro64-6198, SCH-221510, and related compounds from the ChEMBL database—were decomposed into their core fragment components using RDKit's molecular fragmentation tools. The resulting functional group fragments most commonly associated with NOP activity, particularly those appended to the nitrogen-bearing scaffold positions, were designated as 'reference tails.'

In the second stage, approximately 50,000 fragments from the ZINC-22 database—a freely accessible multi-billion-scale database of commercially available compounds—were screened against these reference tails using a weighted similarity scoring function:

$$S_{\text{Final}} = (S_{3D})^{0.4} \times (S_{QM})^{0.30} \times (S_{\text{Phys}})^{0.20} \times (S_{MCS})^{0.10}$$

This equation integrates four complementary similarity dimensions: three-dimensional shape and electrostatic similarity (S_{3D}), quantum chemical descriptor similarity (S_{QM}), physiochemical property similarity (S_{Phys}), and maximum common subgraph similarity (S_{MCS}). Fragments with a composite similarity score exceeding 0.8 relative to the reference tails were retained in the final fragment pool.

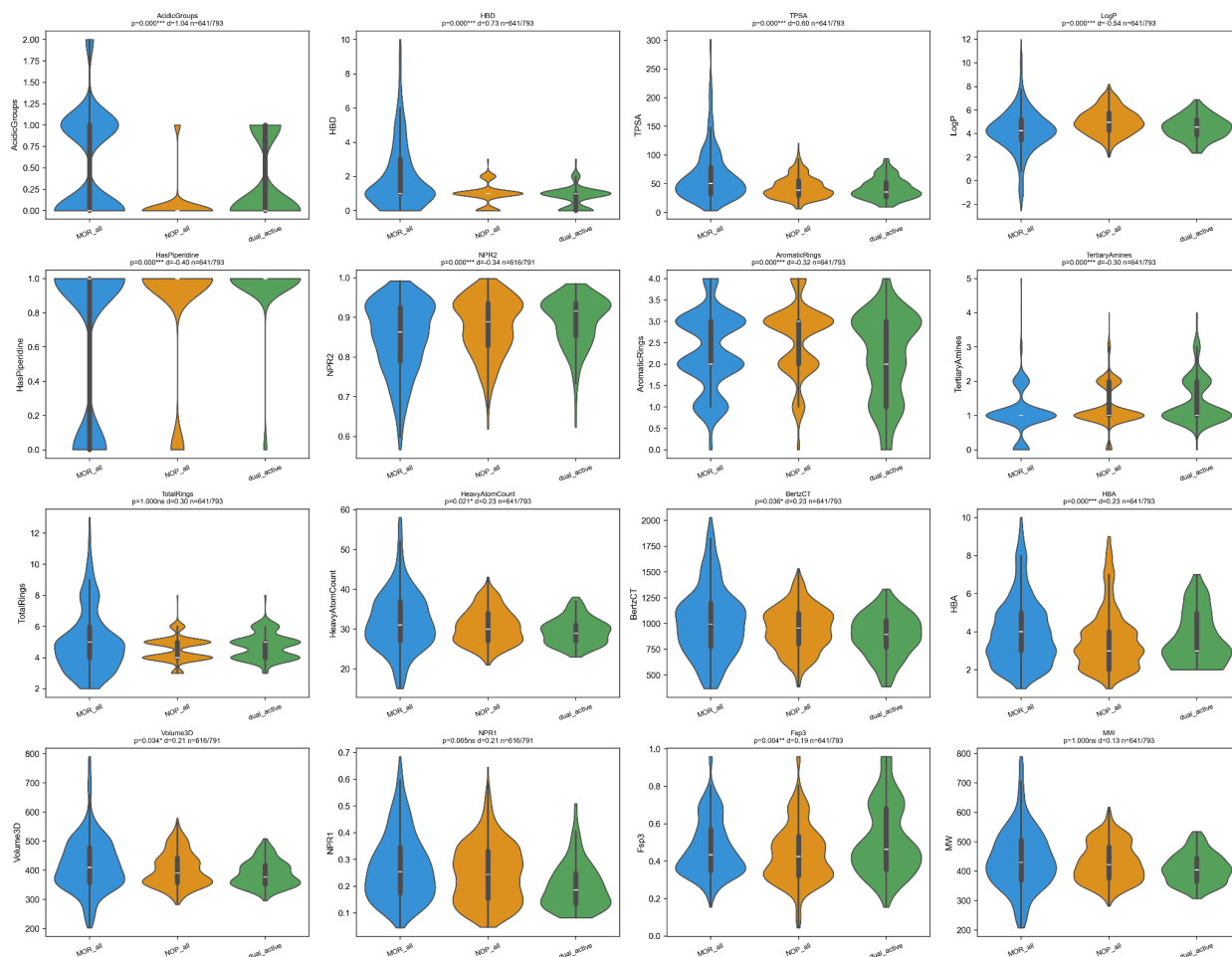


Fig 1: Violin plot for chemical description comparison

Each retained fragment was then coupled to the norfentanyl scaffold in silico to generate candidate analog SMILES strings. These analogs were filtered for drug-likeness using Lipinski's Rule of Five ($MW \leq 500$ Da, $\text{LogP} \leq 5$, $\text{HBD} \leq 5$, $\text{HBA} \leq 10$), as well as additional ADMET-relevant physicochemical filters including topological polar surface area ($\text{TPSA} < 90 \text{ \AA}^2$), aromaticity, and the presence of a basic nitrogen. Structures passing all filters were converted to 3D conformers and saved in SDF format using RDKit's ETKDGV3 embedding algorithm.

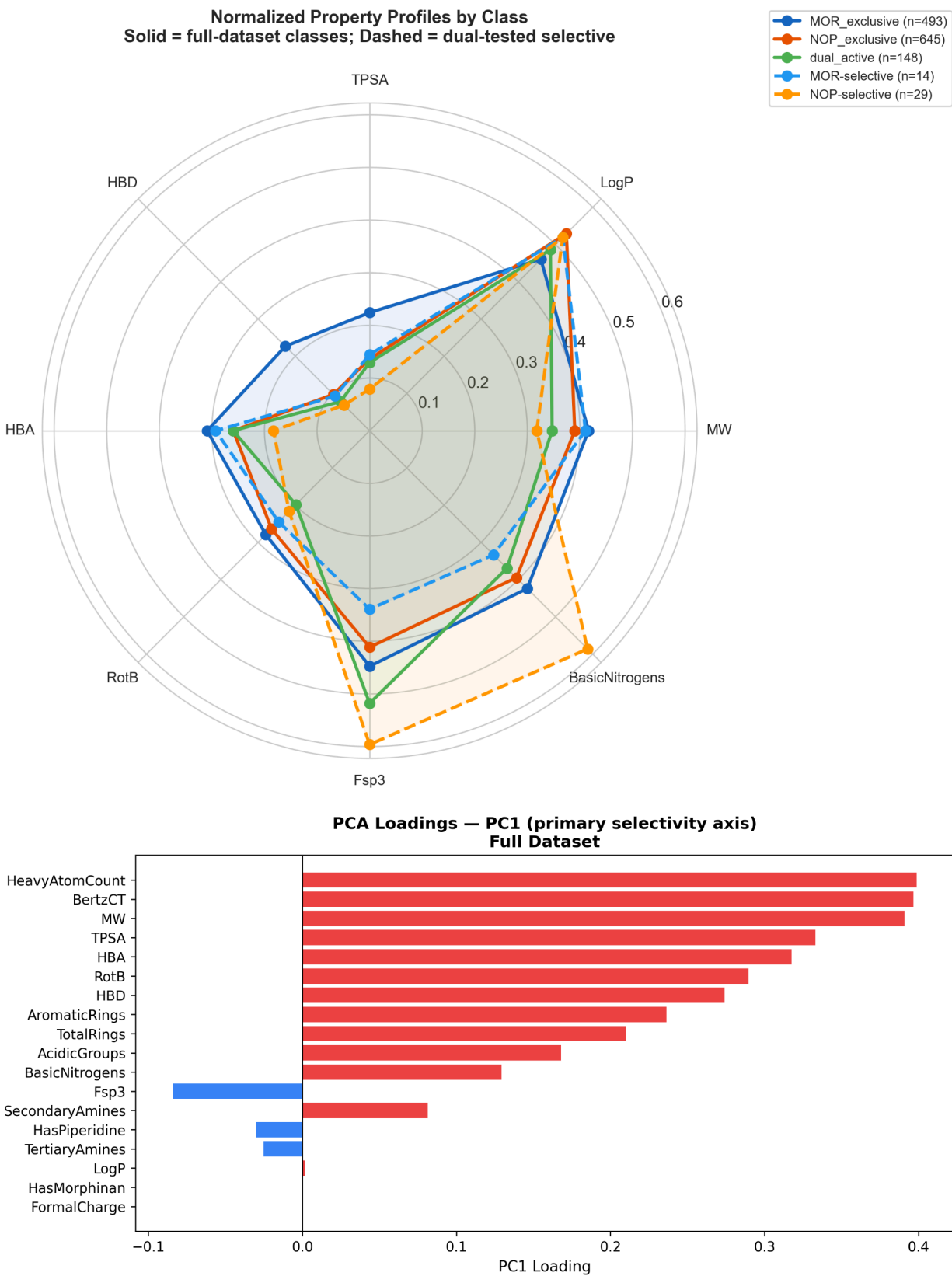


Fig 2: ADMET and Physicochemical Filtering Criteria

Target Protein Preparation

Two receptor protein structures were obtained from the RCSB Protein Data Bank: PDB ID 5C1M (MOR in complex with the agonist BU72) and PDB ID 5DHH (NOP in complex with the antagonist SB-612111). Both structures were processed in UCSF ChimeraX to prepare clean apo-receptor files suitable for docking.

For each structure, the preparation procedure followed four steps: (1) the coordinates of the active site were recorded (MOR: X=1.29, Y=16.45, Z=-59.11 Å; NOP: X=-9.23, Y=29.04, Z=-2.71 Å); (2) the co-crystallized ligand (BU72 for MOR; SB-612111 for NOP) was deleted from the structure; (3) all non-standard residues, including crystallographic water molecules, ions, and auxiliary ligands that would interfere with docking, were removed; and (4) polar hydrogen atoms were added to the cleaned protein to ensure correct protonation states at physiological pH. The resulting pristine receptor files were exported in PDB format and subsequently converted to PDBQT format using Open Babel for compatibility with the AutoDock Vina docking engine.

Molecular Docking

All candidate analog structures were protonated at physiological pH using RDKit and converted from SDF to PDBQT format via Open Babel. Molecular docking was performed using AutoDock Vina, implementing a grid box of 20 × 20 × 20 Å centered on the experimentally determined active site coordinates for each receptor.

AutoDock Vina estimates binding affinity by calculating the Gibbs free energy of binding ($\Delta G = \Delta H - T\Delta S$), where more negative values indicate thermodynamically more favorable binding. Each of the ~57 prepared analog conformers was docked independently to both the prepared MOR and NOP structures. Only compounds achieving a binding affinity of $\Delta G < -9.0$ kcal/mol at both receptors simultaneously were retained for further analysis, as this threshold was chosen to reflect potency comparable to or exceeding that of established reference agonists. This dual-receptor filter reduced the candidate library to four scaffolds: biphenyl, naphthyl, indole, and benzimidazole derivatives of norfentanyl.

Molecular Dynamics Simulations

To assess the stability of the four candidate MOR/NOP-compound complexes over time, molecular dynamics (MD) simulations were performed using OpenMM 8. The docked complexes were exported from AutoDock Vina as .xml-compatible coordinate files. Each simulation was run for 10 picoseconds under physiological conditions (310 K, explicit solvent model) using the AMBER14 force field. Stability was quantified by tracking the C α root-mean-square deviation (RMSD) over time—a lower and more stable RMSD trajectory indicates that the ligand remains well-positioned within the binding pocket without inducing large conformational changes in the receptor.

All four compounds showed RMSD values below 3.0 Å throughout the simulation period, confirming that the identified docking poses were stable. The naphthyl-modified scaffold distinguished itself by displaying the lowest RMSD across both MOR and NOP simulations, suggesting superior binding stability at both targets. This compound was therefore selected as the primary synthesis candidate.

Chemical Synthesis of Naphthyl-Norfentanyl

The synthesis of the naphthyl-modified norfentanyl analog, formally designated N-(1-(naphthalen-2-ylmethyl)piperidin-4-yl)-N-phenylpropionamide ($C_{25}H_{28}N_2O$, MW: 372.5 g/mol), was performed at the OHSU Williams Laboratory under appropriate controlled-substance licensing. The synthesis employed a one-pot reductive amination strategy, a mild and efficient method for functionalizing secondary amines.

Norfentanyl was combined with 2-naphthaldehyde in the presence of acetic acid. The aldehyde first underwent nucleophilic attack by the piperidine nitrogen of norfentanyl to form a hemiaminal intermediate. Protonation by acetic acid facilitated dehydration to generate an iminium ion intermediate, which was subsequently reduced using sodium triacetoxyborohydride ($NaBH(OAc)_3$) as a mild hydride source to yield the final secondary amine product. The reaction was conducted in a single vessel under ambient temperature conditions, avoiding the need for protecting groups or harsh reagents.

Synthesis Characterization

Compound identity and purity were confirmed by two orthogonal analytical methods. Liquid chromatography–mass spectrometry (LC-MS) was performed using a C18 reverse-phase column with a gradient mobile phase. The purified compound showed a single dominant peak at a retention time of 4.23 minutes, with the mass spectrum at that retention time showing a primary $[M+H]^+$ ion at $m/z = 373.5$, consistent with the expected molecular weight of 372.5 g/mol. Minor isotope peaks at 374.5 (M+1, 21.3% relative abundance) and 375.5 (M+2, 4.8%) were consistent with the expected natural isotope distribution for a compound containing 25 carbons and 2 nitrogens.

Proton nuclear magnetic resonance (1H -NMR) spectroscopy was performed in deuterated chloroform. The spectrum confirmed the expected structural features: 12 aromatic protons between 7.0 and 8.5 ppm corresponding to the naphthyl (7H) and phenyl (5H) ring systems, a 2H singlet at 3.70 ppm attributable to the N-CH₂ methylene linker connecting the piperidine nitrogen to the naphthyl group, and the characteristic propionyl group signals consisting of a 2H quartet near 2.3 ppm and a 3H triplet near 1.0 ppm. Integration and multiplicity patterns were fully consistent with the proposed structure. LC-MS purity analysis indicated a purity of 94.8%.

In-Vitro cAMP Signaling Assays

Functional receptor activation was assessed using an HTRF-based cAMP inhibition assay. In this assay format, cells are pre-stimulated with forskolin to elevate intracellular cAMP, generating a fluorescent signal. Activation of Gi-coupled receptors such as MOR or NOP suppresses adenylyl cyclase activity and reduces cAMP levels, dimming the signal. The degree of cAMP suppression is therefore directly proportional to receptor activation, and dose-response curves yield IC₅₀ values reflecting the compound's functional potency.

Three assay configurations were used. First, individual receptor assays were run in HEK293 cells stably expressing either MOR (with Fentanyl as the MOR-selective control agonist) or NOP (with N/OFQ as the NOP-selective control peptide). All four candidate compounds were tested across a concentration range of 0.3 nM to 10,000 nM. IC₅₀ values were extracted by fitting the data to a four-parameter logistic curve. Second, a dual co-expression assay was performed in HEK293 cells stably co-expressing both MOR and NOP. Naphthyl-Norfentanyl was tested alone, alongside fentanyl as a reference MOR-only agonist, and in the presence of the NOP-selective antagonist SB-612111 to pharmacologically isolate the NOP contribution to the observed signal. Maximal inhibition (E_{max}) values from the dual assay were compared across conditions to quantify the degree of NOP-mediated modulation of MOR signaling.

Electrophysiology: Mouse-derived Neuronal GIRK Assay

To obtain a direct neurophysiological measure of receptor-mediated signaling most relevant to respiratory depression risk, ex vivo brain slice electrophysiology was performed at OHSU. cultured primary mouse neurons, which express high levels of MOR and NOP and are a well-established model for opioid electrophysiological studies, were targeted in acute mouse brain slices.

G-protein-coupled inwardly rectifying potassium (GIRK) channel currents were measured as a readout of receptor activation using manual whole-cell patch clamp configuration. GIRK channels open downstream of Gi-protein activation, producing an inwardly rectifying potassium current that can be directly quantified as peak amplitude in picoamperes (pA). Both fentanyl and Naphthyl-Norfentanyl were applied at saturating concentrations, and peak GIRK current amplitudes were recorded under control conditions and in the presence of the adenylyl cyclase activator forskolin. Peak current amplitudes were compared between the two compounds, and statistical significance was assessed using a two-sample comparison with significance set at $p < 0.001$.

Results

Computational Docking and Fragment Selection

The fragment-based screening pipeline identified 57 norfentanyl-derived analog structures that passed all drug-likeness and ADMET filters. Following dual-receptor docking with the -9.0 kcal/mol threshold, four scaffolds qualified: biphenyl, naphthyl, indole, and benzimidazole. Their respective binding affinities are summarized in the table below.

Scaffold	MOR ΔG (kcal/mol)	NOP ΔG (kcal/mol)	Comparative NOP Ref. (Ro64-6198)
Biphenyl	-10.1	-8.3	
Naphthyl	-9.8	-9.2	Selected for synthesis
Benzimidazole	-9.4	-8.7	
Indole	-9.5	-9.0	
Original Fentanyl (ref.)	-11.2	—	MOR reference only
Ro64-6198 (ref.)	—	-9.1	NOP reference

Table 1. Binding affinities of four top docking candidates at MOR and NOP, compared to reference compounds. All values represent the top-ranked AutoDock Vina pose. More negative ΔG values indicate stronger predicted binding.

Molecular Dynamics Stability

All four candidate compounds maintained RMSD values below 3.0 Å over the 10 ps simulation window at both MOR and NOP, confirming overall binding stability. The naphthyl scaffold was clearly distinguished from the other three candidates by its consistently low RMSD trajectory. At

MOR, the naphthyl compound reached a plateau of approximately 1.0 nm RMSD, substantially lower than biphenyl (~1.5 nm), indole (~1.75 nm), and benzimidazole (~2.75 nm). At NOP, the naphthyl analog showed even more stable binding, plateauing near 0.4 nm, compared to biphenyl (~0.6 nm), indole (~0.7 nm), and benzimidazole (~0.9–1.0 nm). The consistent superiority of the naphthyl scaffold across both target simulations provided strong computational justification for its prioritization in synthesis.

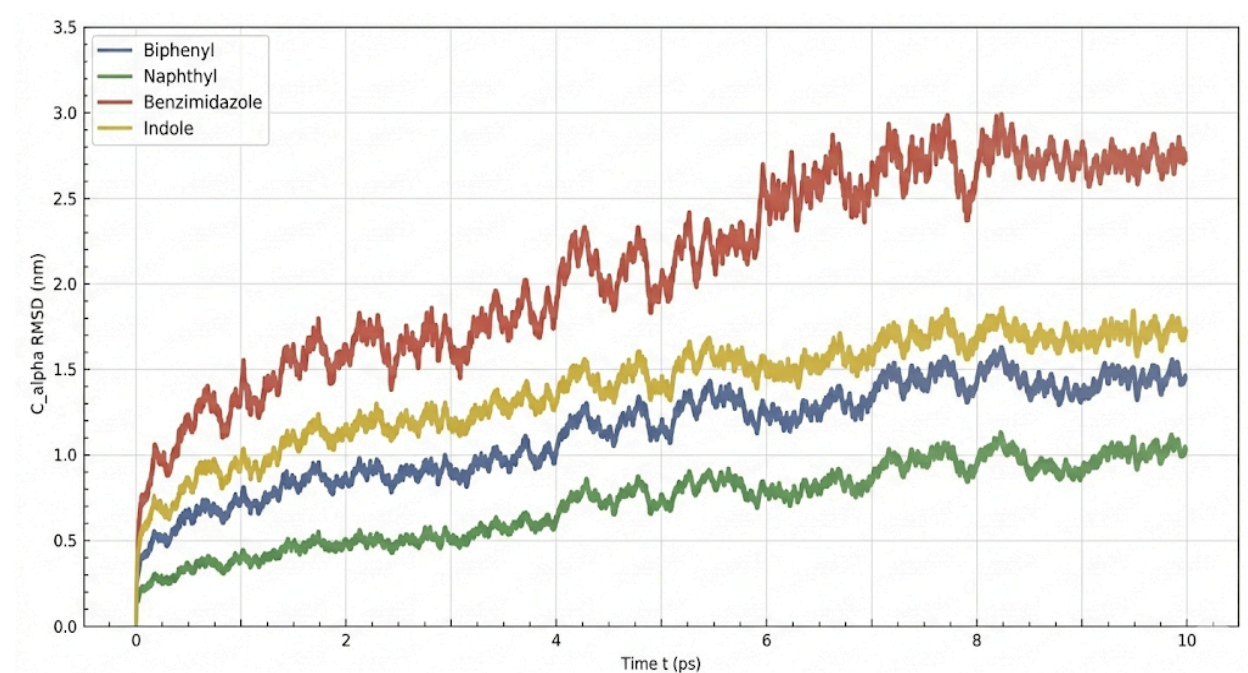


Fig 3: RMSD results for top 4 compound candidates against MOR Receptor

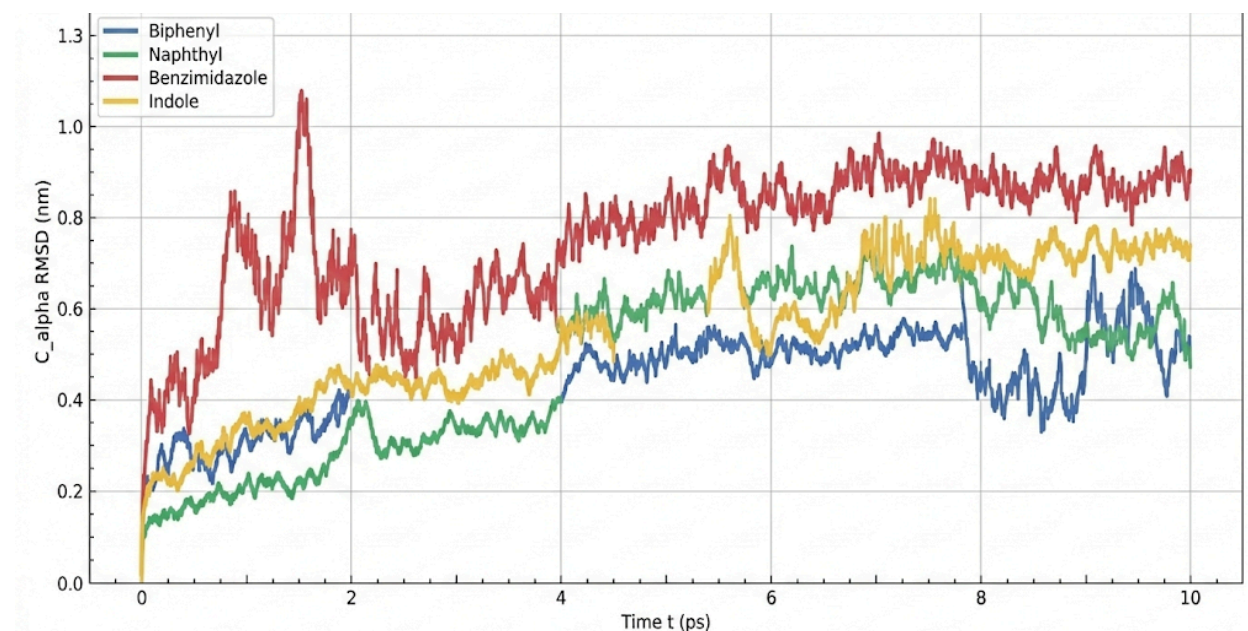
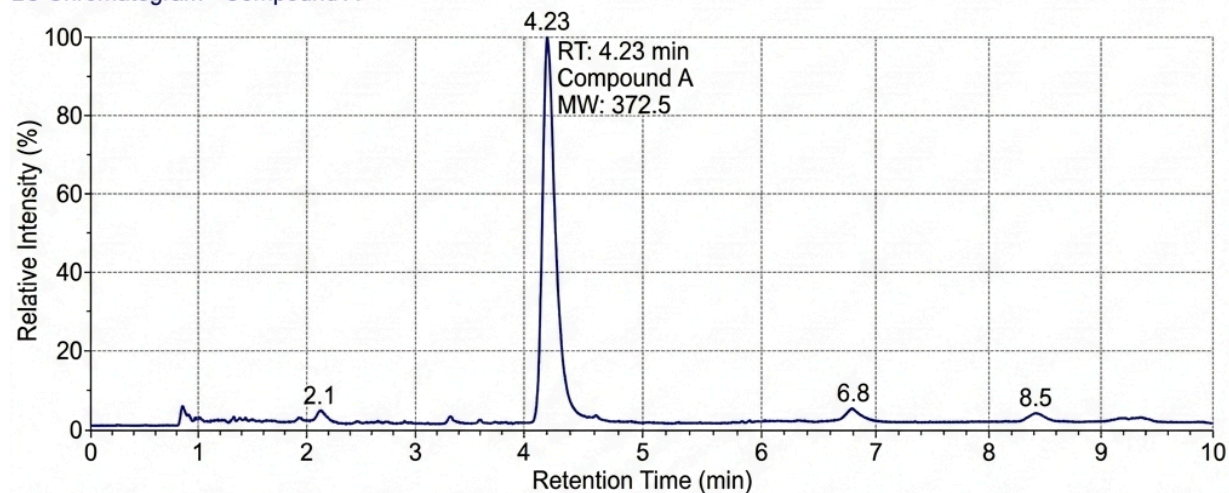


Fig 4: RMSD Results for top 4 compound candidates against NOP Receptor

Synthesis and Characterization

Naphthyl-Norfentanyl was successfully synthesized via the described one-pot reductive amination. LC-MS confirmed a dominant peak at retention time 4.23 minutes with the expected $[M+H]^+ = 373.5$. $^1\text{H-NMR}$ spectral analysis confirmed all expected structural features, including the 12-proton aromatic cluster, the methylene singlet, and the propionyl quartet and triplet. Purity was determined to be 94.8% by peak area integration of the LC chromatogram, confirming the synthesis had proceeded cleanly with minimal byproducts and that the compound was suitable for direct in-vitro pharmacological testing without additional purification.

LC Chromatogram - Compound A



Mass Spectrum - RT 4.23 min

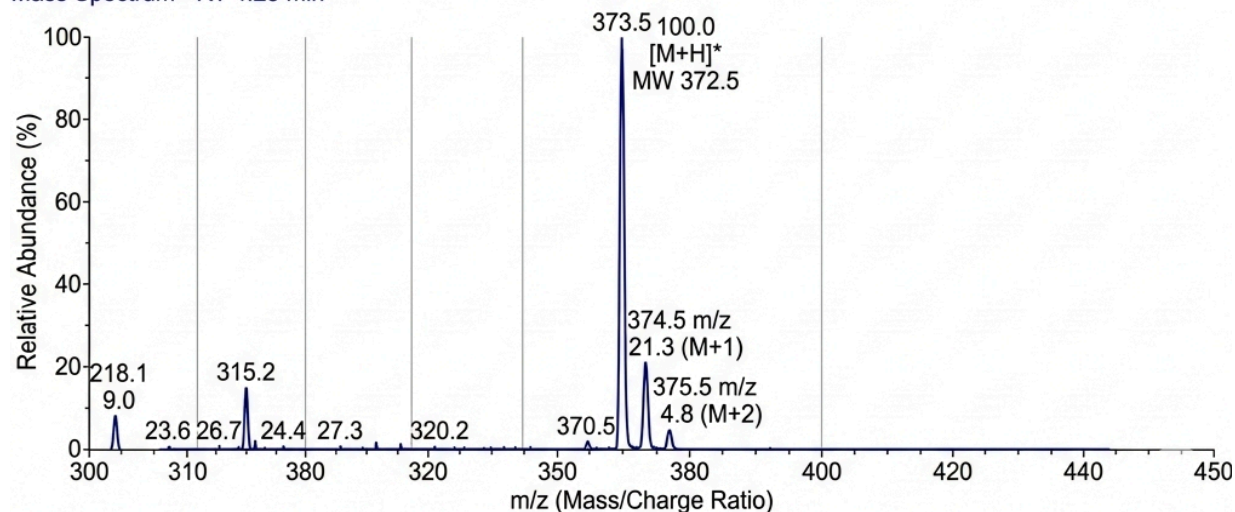


Fig 5: Liquid Chromatography-Mass Spectrometry Results

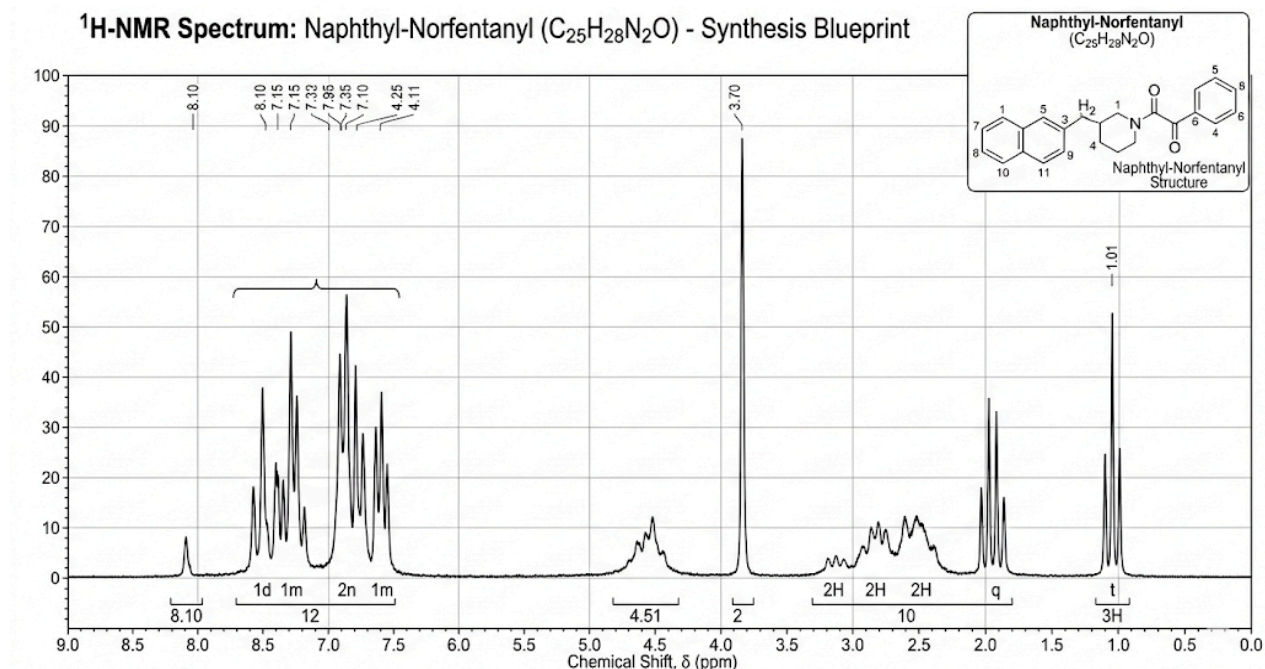


Fig 6: Proton-Nuclear Magnetic Resonance Results

Individual cAMP Assays: MOR and NOP Potency

In HEK293 cells expressing MOR alone, Naphthyl-Norfentanyl produced concentration-dependent cyclic AMP (cAMP) inhibition with a half-maximal inhibitory concentration (IC₅₀) of approximately 5 nM. For comparison, fentanyl—the gold-standard MOR agonist—yielded an IC₅₀ of 3 nM under identical conditions. The modest 1.7-fold difference indicates that the naphthyl modification does not substantially diminish MOR potency, confirming that the compound retains near-fentanyl analgesic potency at the target receptor.

In HEK293 cells expressing NOP alone, Naphthyl-Norfentanyl produced an IC₅₀ of 11 nM, demonstrating robust functional NOP activation. This value is significantly more potent than the reference NOP agonist Ro64-6198, which produced an IC₅₀ of approximately 20 nM under the same conditions. These results confirm that the naphthyl modification successfully confers genuine NOP activity while preserving MOR activity—meeting the core design objective of bifunctional dual agonism.

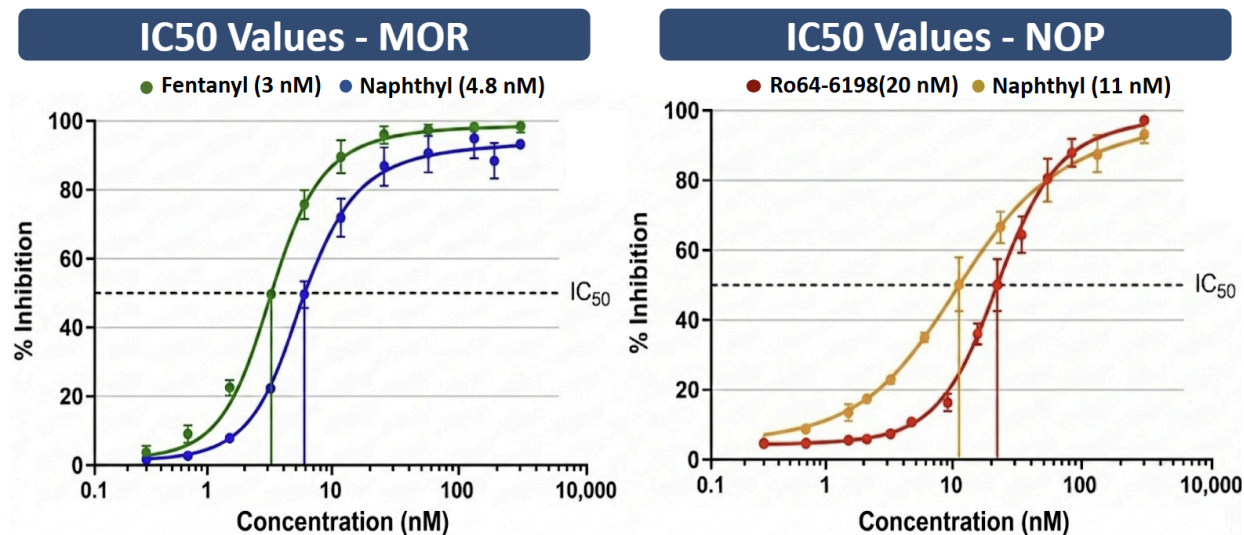


Fig 7: cAMP IC₅₀ results for MOR and NOP Receptors compared to reference agonists

Dual Co-Expression cAMP Assay: NOP Modulation of MOR Signaling

The most functionally significant results emerged from the dual MOR/NOP co-expression assay. In cells expressing both receptors, the E_{max} of cAMP inhibition by fentanyl was 93%, consistent with near-maximal G_i activation through MOR. In contrast, Naphthyl-Norfentanyl produced an E_{max} of only 55% in the same co-expressing system—a 38-percentage-point reduction. To confirm this reduction was specifically attributable to concurrent NOP engagement, the NOP antagonist SB-612111 was added to block NOP while allowing MOR to remain active. Under these NOP-blocked conditions, the E_{max} of Naphthyl-Norfentanyl recovered to 87%, a level much closer to fentanyl's uninhibited MOR-only ceiling.

This pharmacological dissection provides strong evidence that NOP co-activation by the naphthyl analog is actively and substantially counteracting MOR-driven signaling in real time. The shift in E_{max} from 87% (NOP blocked) to 55% (NOP active) represents a 32-percentage-point reduction attributable specifically to NOP engagement. Given that G_i-coupled receptor signaling in the brainstem respiratory centers is the primary mechanistic driver of opioid-induced respiratory depression, a compound that inherently limits its own maximal G_i signaling ceiling via dual-receptor engagement represents a fundamentally different and potentially safer pharmacological profile.

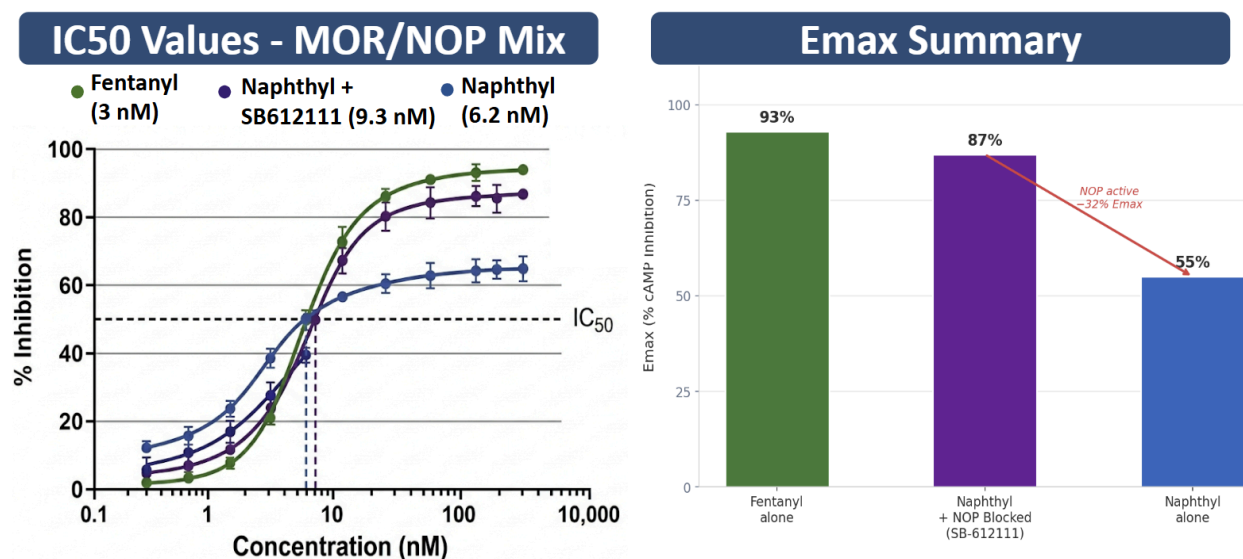


Fig 8: cAMP IC₅₀ results and summary for IC₅₀ values in dual-expression HEK293 Cells

Discussion

Interpretation of Computational Results

The docking and molecular dynamics results suggest that the naphthyl modification of norfentanyl's piperidine nitrogen achieves its dual-receptor profile through distinct structural interactions at each target. The naphthyl group's bicyclic aromatic ring system likely forms additional hydrophobic contacts within the NOP binding pocket compared to fentanyl's monoaromatic phenethyl group, while the overall scaffold dimensions remain compatible with the MOR binding site. The superior MD stability of the naphthyl compound at both receptors relative to the benzimidazole and indole analogs—which showed higher RMSDs, particularly at MOR—supports the hypothesis that ring geometry and lipophilicity are critical determinants of dual-receptor binding stability.

It should be noted that AutoDock Vina's scoring function, while computationally efficient, is a semi-empirical approximation. The 10 ps MD simulation window used in this study is relatively short compared to the microsecond-to-millisecond timescales of physiologically relevant conformational changes. These computational results are therefore best interpreted as hypothesis-generating tools that guide compound selection, rather than definitive predictors of in-vivo activity. The subsequent experimental validation provides the primary evidence base for conclusions about this compound's pharmacological profile.

Alignment with Published Dual NOP/MOR Pharmacology

The in-vitro results obtained for Naphthyl-Norfentanyl are broadly consistent with the growing body of literature on dual NOP/MOP agonism. Ciccocioppo et al. (2025) reviewed clinical and preclinical data supporting the hypothesis that NOP activation attenuates MOR-mediated respiratory depression through Gi signaling cross-regulation in brainstem circuits. Cebanopadol, the most clinically advanced NOP/MOP dual agonist, has demonstrated analgesic efficacy in Phase II clinical trials with a reportedly improved respiratory safety margin, lending translational credibility to the bifunctional agonism approach pursued here.

The 55% Emax observed for Naphthyl-Norfentanyl in the dual co-expression system is notably lower than what has been reported for cebranopadol (which retains somewhat higher Emax values at MOR), suggesting that the naphthyl compound may in fact produce more pronounced NOP-mediated MOR suppression. This could reflect differences in the relative MOR:NOP potency ratio, receptor expression levels in the assay system, or intrinsic efficacy differences at NOP. Further comparison with cebranopadol under identical assay conditions would clarify this distinction.

Potential Sources of Error and Limitations

Several limitations of the current study should be acknowledged. First, all in-vitro assays were performed in heterologous HEK293 expression systems and ex vivo mouse neurons. While these are well-validated pharmacological models, they do not capture the full complexity of receptor expression levels, accessory proteins, and circuit-level interactions in the intact CNS. Second, the cAMP assay measures Gi-mediated signaling specifically and does not capture potential beta-arrestin recruitment, which is also relevant to opioid side-effects including tolerance and constipation. Third, the study did not include radioligand binding assays (e.g., competitive displacement with [³H]-Fentanyl or [³H]-N/OFQ), which would provide direct measurements of binding affinity (K_i) independent of functional readouts. Finally, pharmacokinetic properties—including CNS penetration, metabolic stability, and plasma half-life—have not yet been characterized, and these will be essential for translational development.

It is also important to note that this compound, as a fentanyl derivative, falls under Schedule I/II controlled substance regulations. All synthesis was performed under appropriate DEA-licensed supervision at OHSU, and any further development would require full regulatory engagement. The compound is not being proposed as an immediately deployable drug, but rather as a validated proof-of-concept scaffold demonstrating that bifunctional MOR/NOP agonism is achievable through rational modification of the norfentanyl scaffold.

Future Directions

Several near-term experimental priorities are identified. Radioligand binding assays using tritiated reference ligands should be conducted to determine binding affinity (K_i) values at both receptors independently of functional coupling. The remaining three docking candidates (biphenyl, indole, benzimidazole analogs) should be synthesized and pharmacologically characterized to establish a comprehensive structure-activity relationship (SAR) across the scaffold series. Beta-arrestin recruitment assays (e.g., BRET-based) would provide a more complete picture of biased agonism profiles.

Longer-term, iPSC-derived human neurons—particularly preBötC-like respiratory neurons—would provide a more disease-relevant cellular model for evaluating respiratory safety. In vivo pharmacological profiling in rodent models should include standard antinociceptive assays (hot plate, tail flick), quantitative plethysmography for respiratory rate measurements, and conditioned place preference assays to assess addiction potential. These studies would provide the translational evidence necessary to evaluate this compound class as a legitimate candidate for further pharmaceutical development.

Conclusions

This project successfully demonstrated that rational modification of the norfentanyl scaffold via substitution at the piperidine nitrogen can produce bifunctional MOR/NOP dual agonists with a substantially improved pharmacological safety profile. The naphthyl-modified analog, Naphthyl-Norfentanyl, was computationally identified through a rigorous multi-parameter screening workflow, synthesized at 94.8% purity via reductive amination, and validated through three independent in-vitro experimental platforms.

The compound achieved an IC₅₀ of 5 nM at MOR—closely comparable to fentanyl's 3 nM—while simultaneously achieving an IC₅₀ of 11 nM at NOP, outperforming the reference NOP agonist Ro64-6198 at 20 nM. In dual co-expression systems, NOP co-activation reduced the MOR signaling ceiling by 38 percentage points (from 87% to 55% E_{max}), confirming active real-time NOP-mediated attenuation of MOR signaling. Neuronal GIRK electrophysiology demonstrated approximately 2-fold lower peak currents for Naphthyl-Norfentanyl versus fentanyl ($p < 0.001$), providing direct physiological evidence of reduced receptor-driven signaling in native neural tissue.

Collectively, these results support the central hypothesis that bifunctional MOR/NOP agonism can preserve analgesic potency while intrinsically limiting the ceiling of MOR-driven side effects. This work contributes a validated proof-of-concept scaffold and a reproducible cheminformatic design pipeline that could be applied more broadly to the systematic development of safer opioids. The potential applications extend from safer prescription pain management for chronic pain and surgical analgesia to low-monitoring analgesic options in resource-limited settings such as remote or military field environments. Addressing the opioid crisis ultimately requires not just better overdose treatment, but better drugs—and this project represents a step toward designing opioids that are intrinsically safer by pharmacological design.

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